Rahul Dange

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Technical Skills

Systems & Low-Level: C, C++, x86-64 Assembly, Linux Kernel, POSIX Threads, Concurrency, IPC, Memory Management, GDB, Valgrind

Languages & Paradigms: Python, Java, TypeScript, JavaScript, SQL (PostgreSQL), Microservices Architecture, System Design, SOLID

Frameworks & APIs: React, Next.js, Node.js, Express, FastAPI, RESTful Architecture

Infrastructure & Telemetry: Linux, Bash/Shell, Git, Docker, Google Cloud (GCP), AWS, CI/CD (GitHub Actions), OpenTelemetry

Experience

Academic Tutor – COMP2017 : System Programming (Part-Time) Sydney, Australia
The University of Sydney *Feb 2026 – Present*

- Mentoring 200+ students in low-level system programming (C/C++), debugging volatile memory and concurrency faults via GDB/Valgrind.
- Managing EdStem query workflows and administrative tracking, strictly categorized under the GEN-EA CLAUSE 68 compliance framework.

Full Stack Engineer (Part-Time) Pune, India
ChessVerst *Feb 2021 – Jul 2023*

- Architected scalable software solutions and full-stack features utilizing JavaScript (ES6+), TypeScript, and modern architectural patterns.
- Engineered decoupled backend microservices to optimize data transport protocols and ensure high-availability cross-platform performance.

Associate Software Engineer (Part-Time) Remote
Calc Technologies *Mar 2022 – May 2022*

- Developed dynamic, responsive user interfaces leveraging React.js, enhancing platform usability and client-side rendering speeds.

Technical Projects

Geometric Storage Database | ***ISO C11, Direct OS I/O, Advanced Math***

- Engineered an experimental $\mathcal{O}(1)$ constant-time document store, bypassing traditional B-Trees by mapping 4D mathematical coordinates directly to physical disk offsets.

Sentinel Daemon and Client CLI | ***C, Assembly, macOS/Linux Internals***

- Developed a zero-dependency autonomous daemon executing continuous malware sweeps, memory intrusion auto-healing, and proactive RAM purging.
- Engineered dynamic background CPU throttling mechanisms to optimally prioritize and scale interactive hardware workloads.

SpatialPress – Binary Compression Codec | ***Algorithms, 64-bit Range Coding***

- Architected a high-performance, modular binary compression codec utilizing multi-dimensional coordinate interleaving via 4D Morton Z-order curves.
- Integrated 64-bit adaptive range coding algorithms to maximize spatial data compression density and decoding throughput.

Spotlight – Social Interaction Platform | ***React, Node.js, Express, MongoDB, Lerna***

- Built a scalable, full-stack interaction platform structured entirely within a centralized monorepo architecture for streamlined deployment pipelines.
- Engineered dynamic client interfaces with JavaScript/React while orchestrating asynchronous backend data transport using Node.js and MongoDB.

Education

The University of Sydney — Sydney, Australia
Master of Computer Science (Specialization in Software Engineering) — *Feb 2024 – Dec 2025*

Nutan College of Engineering and Research (NCER) — Pune, India
Bachelor of Technology in Computer Science and Engineering (Affiliated with DBATU) — *Aug 2019 – Aug 2023*

Publications

Revolutionizing Student Engagement: Building a Cutting-Edge Interaction System — Nov 2023
SSRN (Elsevier) | 5th Intl. Conf. on Communication & Information Processing (ICCIP-2023)

- Authored a technical blueprint for a high-performance interaction platform utilizing the MERN stack and Next.js, analyzing Server-Side Rendering (SSR) to mitigate traditional MVC performance bottlenecks.
- Evaluated system response times under heavy concurrent loads, benchmarking optimized asset delivery pipelines and state management using TypeScript, Cloudinary, and Turbopack.

Student Interaction System Using Modern Web Technologies — Feb 2023
Scandinavian Journal of Information Systems, 35(1), 142–151

- Co-authored research analyzing the integration and architectural implementation of modern web frameworks to optimize educational interaction platforms.
- Published findings on system scalability, web technology utilization, and platform performance in collaboration with academic researchers.

Volunteering & Leadership

Senior Student Advisor — Pune, India
Nutan College of Engineering and Research (NCER) — *Jul 2022 – Jul 2023*

- Appointed by the Head of the CS Department (Dr. Sanjeevkumar Angadi) as a strategic mentor to facilitate the professional transition of junior cohorts into software engineering careers.
- Served as the primary administrative liaison between the student body and faculty leadership to proactively resolve academic roadblocks.

Primary Student Coordinator, NCER Hackathon 2022 — Pune, India
Nutan College of Engineering and Research (NCER) — *2022*

- Spearheaded the institution's inaugural large-scale hackathon as the sole coordinator, representing and managing students across all four years of the CSE department.
- Provided rigorous technical mentorship to 8 cross-functional teams, guiding them through product architecture, technical roadblocks, and execution strategies.

NCER++ LMS Maintainer — Pune, India
Nutan College of Engineering and Research (NCER) — *Feb 2020 – Sep 2023*

- Engineered and scaled a foundational Java application into a comprehensive Learning Management System, delivering academic resources and unit-specific technical roadmaps.
- Transformed the platform into a centralized community hub to equip students with career-building information and industry-ready skills.

Founder & Coordinator, NCER Cyber Security Club — Pune, India
Nutan College of Engineering and Research (NCER) — *Sep 2019 – Sep 2022*

- Established the institution's first cybersecurity community focused on ethical hacking, digital safety, and hands-on application of real-world security concepts.
- Launched and managed the official NCER Cybersecurity Blog to disseminate technical write-ups and security research to the broader student body.

Batch Representative, English Proficiency Initiative — Pune, India
Nutan College of Engineering and Research (NCER) — *Aug 2019 – May 2020*

- Collaborated directly with institutional faculty to optimize teaching methodologies, successfully tailoring the curriculum for students from regional backgrounds.
- Mentored peers to bridge communication gaps, ensuring language proficiency was utilized effectively for professional career development and corporate placements.